Biological Databases in Computational Biology

Course Instructor: Chioma Onyido

Learning Objectives

- Define what biological databases are and explain their role in computational biology research.
- Differentiate between primary and secondary databases with relevant examples.
- Navigate and interpret data from three major databases: TCGA, Ensembl and UCSC Genome Browser.

Learning Objectives

- Understand the structure of biological database entries.
- Recognize how data from these databases can be integrated into tools and pipelines for various tasks.
- Perform a basic query or exploration using one of the databases (hands-on or demo-based).

What are Biological Databases?

- **Biological databases** help to **store**, **search** and **retrieve** massive biological data so that scientists globally can work efficiently.

What Types of Data Do They Store?

- Genomic Sequences (DNA, RNA)
- Gene Expression Data
- Variants (SNPs, mutations)
- Epigenomic & Regulatory Data
- Clinical Metadata (e.g., patient survival)

Importance of Biological Databases

- Store biological data in **standardized,** accessible ways.
- Enable **reproducibility** and **collaboration**.
- Power tools, pipelines and discoveries.

Types of Biological Databases

- **Primary:** Raw data (e.g., TCGA, ENA, SRA).
- **Secondary:** Curated/annotated (e.g., Ensembl, RefSeq).
- **Specialized:** Focused domains (e.g., miRBase, BugSigDB).

Overview of Key Databases

Database	Focus	Type of Data
TCGA	Cancer genomics	Genomic, transcriptomic, clinical
Ensembl	Comparative genomics	Genes, variants, annotations
UCSC	Visualization & genome tools	Tracks: genes, conservation, custom

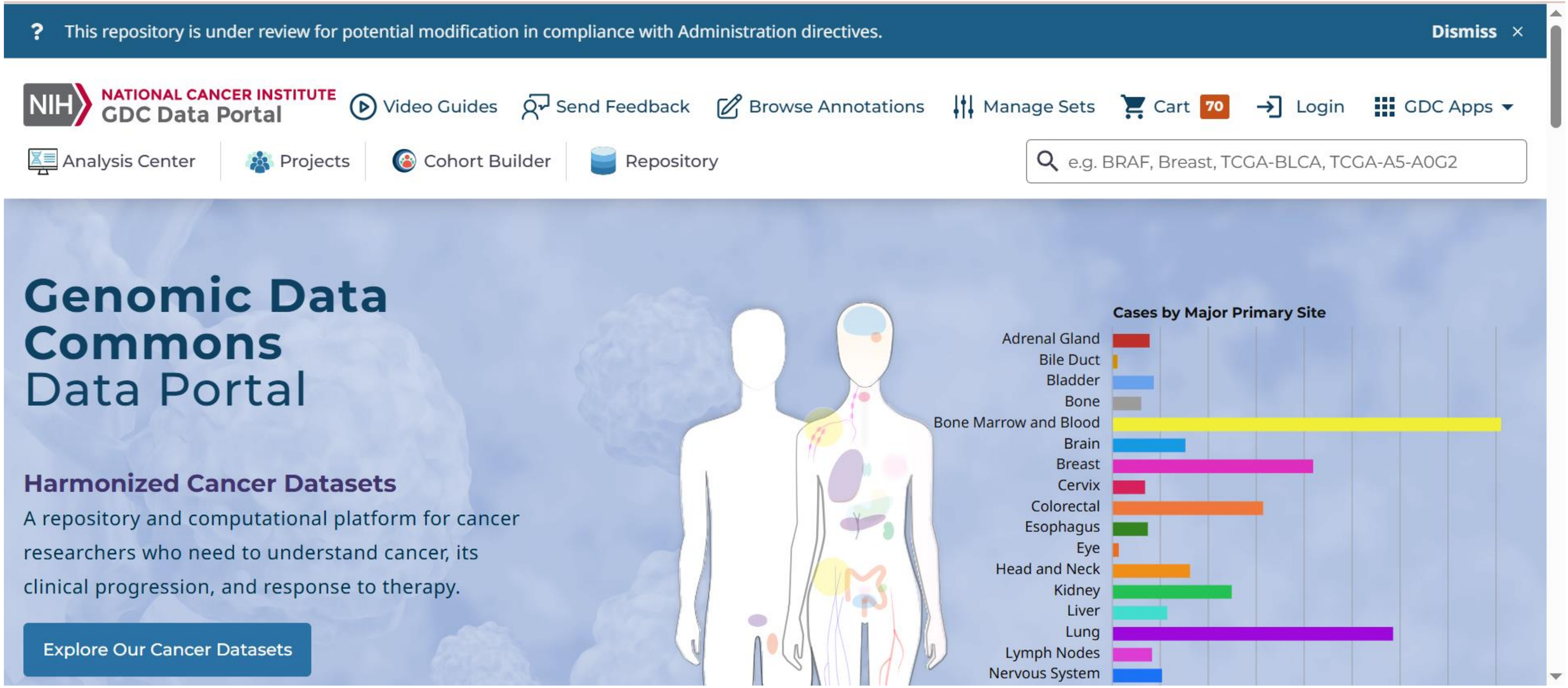
TCGA – The Cancer Genome Atlas

- Hosted by **GDC** (Genomic Data Commons).
- **Multi-omics** cancer database - DNA, RNA, methylation, CNVs, clinical data.
- **11,000** tumor samples across **33** cancer types.
- TCGA is now accessible via the GDC portal, maintained by the NCI.

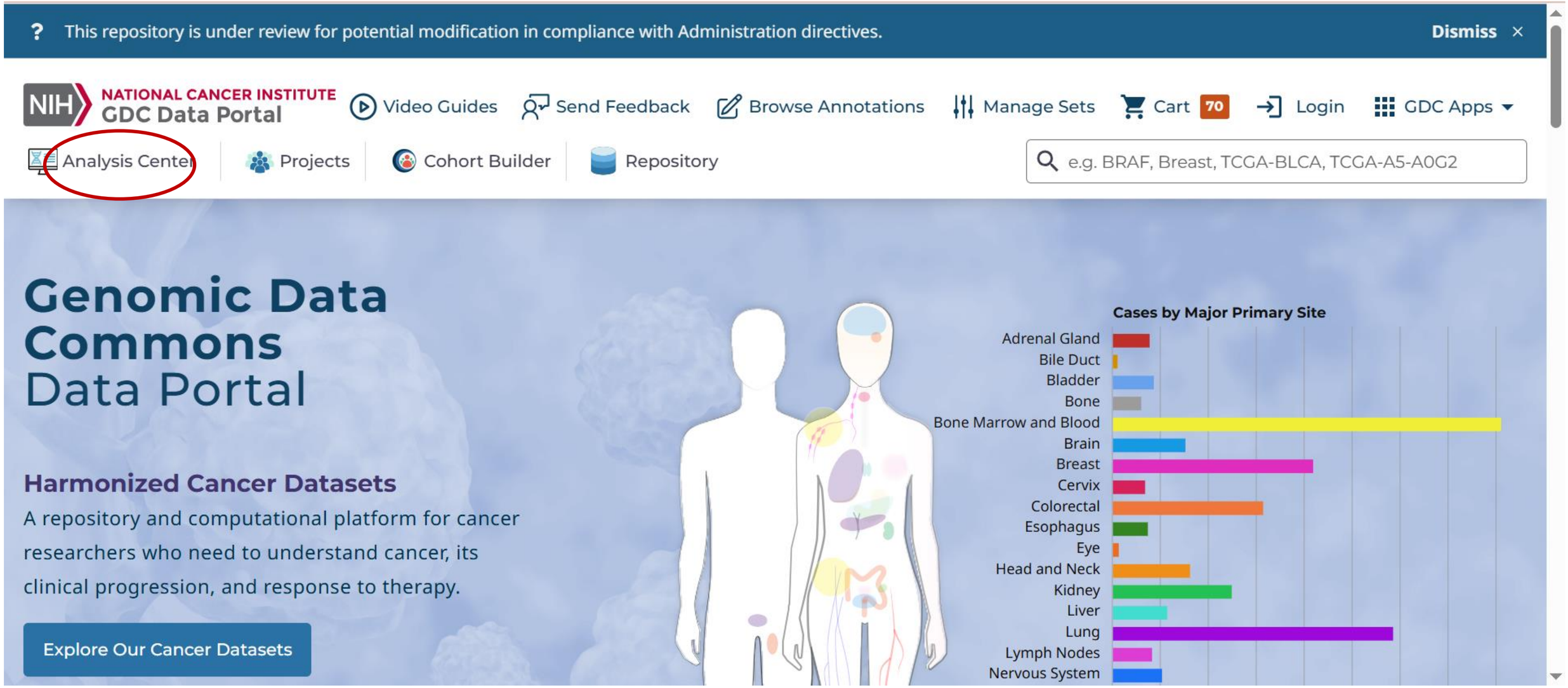
TCGA – How to Access and Use

- **Portal**: https://portal.gdc.cancer.gov
- **Data types**: VCF, CNV, Expression matrices, Clinical metadata
- **Tools**: TCGAbiolinks (R), FireBrowse, Xena Browser

TCGA – Homepage



TCGA – 'Analysis Center'



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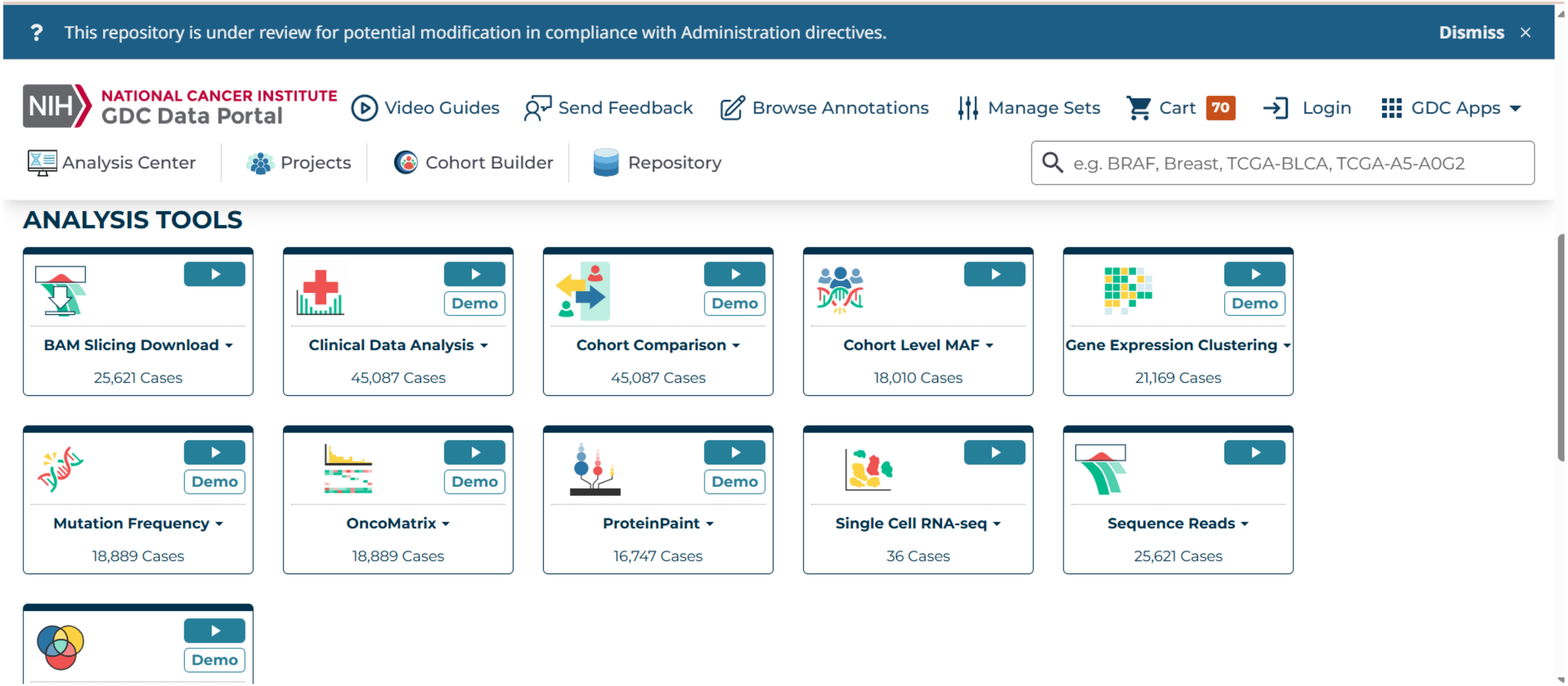
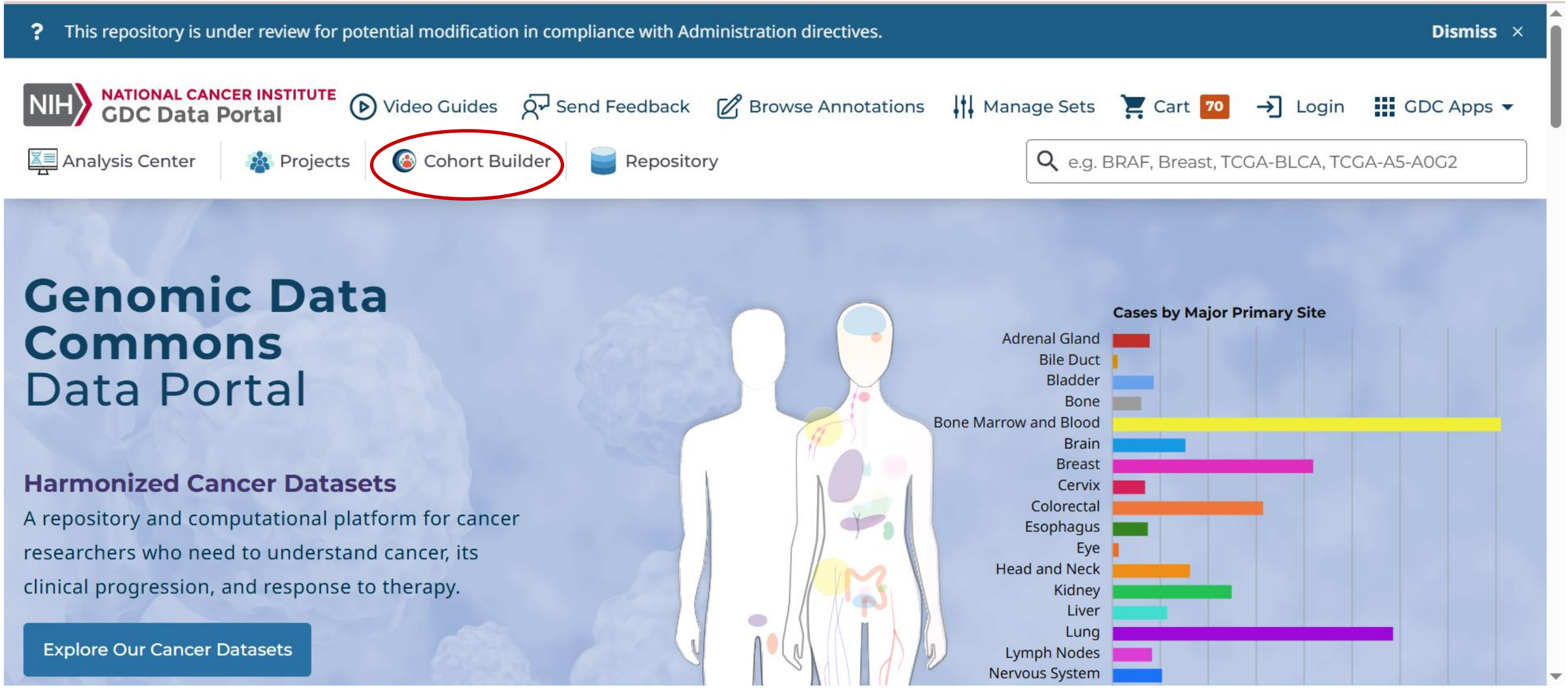


TCGA – 'Cohort Builder'



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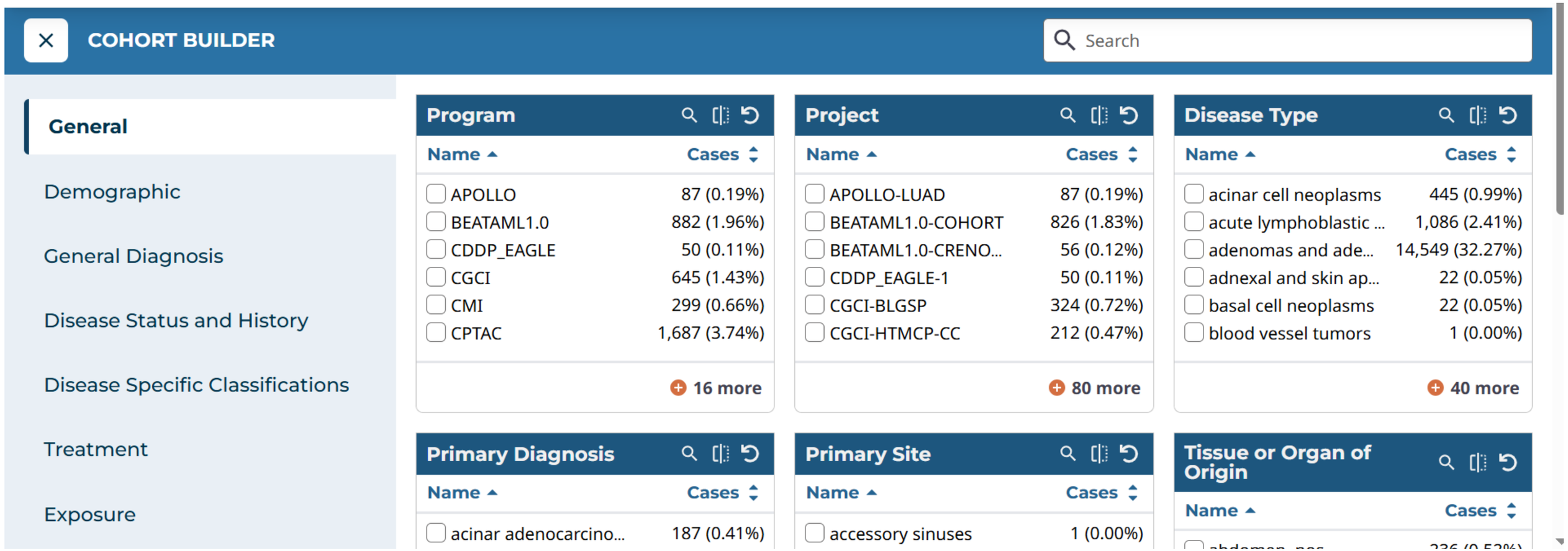
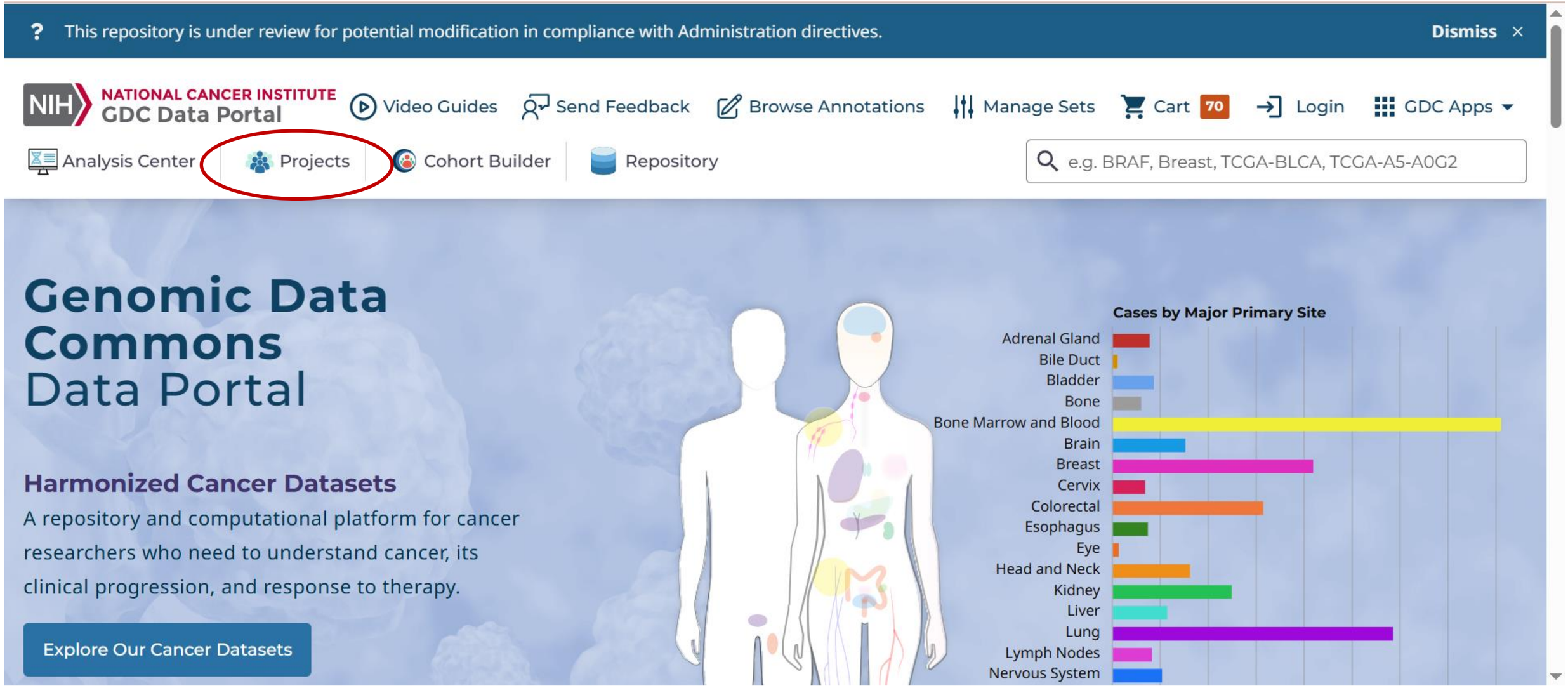


TCGA – 'Projects'



TCGA – 'Projects'

NIH NATIONAL CANCER INSTITUTE GDC Data Portal | Video Guides | Send Feedback | Browse Annotations | Manage Sets | Cart 70 | Login | GDC Apps

Analysis Center | Projects | Cohort Builder | Repository | e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

PROJECTS

Filters

Collapse All

Primary Site

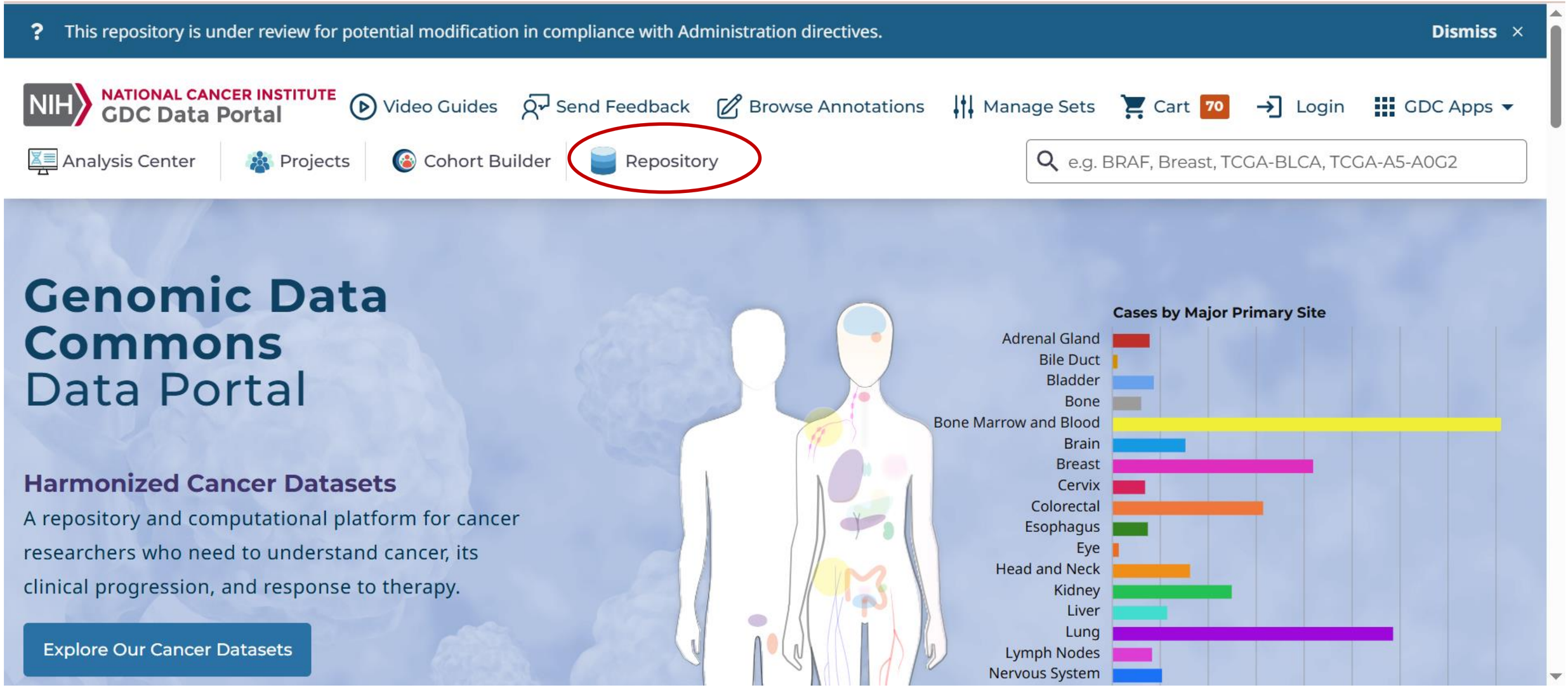
Name	Projects
accessory sinuses	1 (1.16%)
adrenal gland	7 (8.14%)
anus and anal canal	8 (9.30%)
base of tongue	4 (4.65%)
bladder	10 (11.63%)
bones, joints and articul...	3 (3.49%)

63 more

Save New Cohort | JSON | TSV | TOTAL OF **86** PROJECTS | Search

Project	Disease Type	Primary Site	Program	Cases	Experimental Strategy
FM-AD	23 Disease Types	42 Primary Sites	FM	18,004	Targeted Sequencing
TARGET-AML	2 Disease Types	2 Primary Sites	TARGET	2,492	Genotyping Array, Methyla Seq, RNA-Seq, Targeted Se
TARGET-ALL-P2	Lymphoid Leukemias	Hematopoietic and reticuloendothelial systems	TARGET	1,587	Genotyping Array, miRNA-WXS
MP2PRT-ALL	2 Disease Types	Hematopoietic and reticuloendothelial systems	MP2PRT	1,510	RNA-Seq, WGS, WXS

TCGA – 'Repository'



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REPOSITORY

Filters

Collapse All

+ Add a Custom Filter

Experimental Strategy

Name	Files
ATAC-Seq	410 (0.03%)
Diagnostic Slide	11,766 (0.99%)
Expression Array	1,426 (0.12%)
Genotyping Array	147,734 (12.42%)
Methylation Array	50,913 (4.28%)
miRNA-Seq	56,067 (4.71%)

Sample Sheet | Metadata | Manifest | View Images | Add All Files to Cart

Remove All From Cart

TOTAL OF **1,189,760** FILES **45,087** CASES **9.53** PB

JSON | TSV | Search

Cart	Access	File Name	Cases	Pro
	Open	450e8639-47ee-4919-8b96-9770d8235a6b_noid_Red.idat	1	H
	Contro...	HCMI-CMDC.1881b759-eeef-4fa7-aa97-4f7b22c95550.arriba.rna_fusion.bedpe	1	H

TCGA – 'Repository'

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miRNA-Seq	56,067 (4.71%)

7 more

Wgs Coverage

Name	Files
0x-10x	4,459 (0.37%)
10x-25x	4,419 (0.37%)
150x+	3 (0.00%)
25x-150x	27,557 (2.32%)

TOTAL OF **1,189,760** FILES **45,087** CASES **9.53** PB

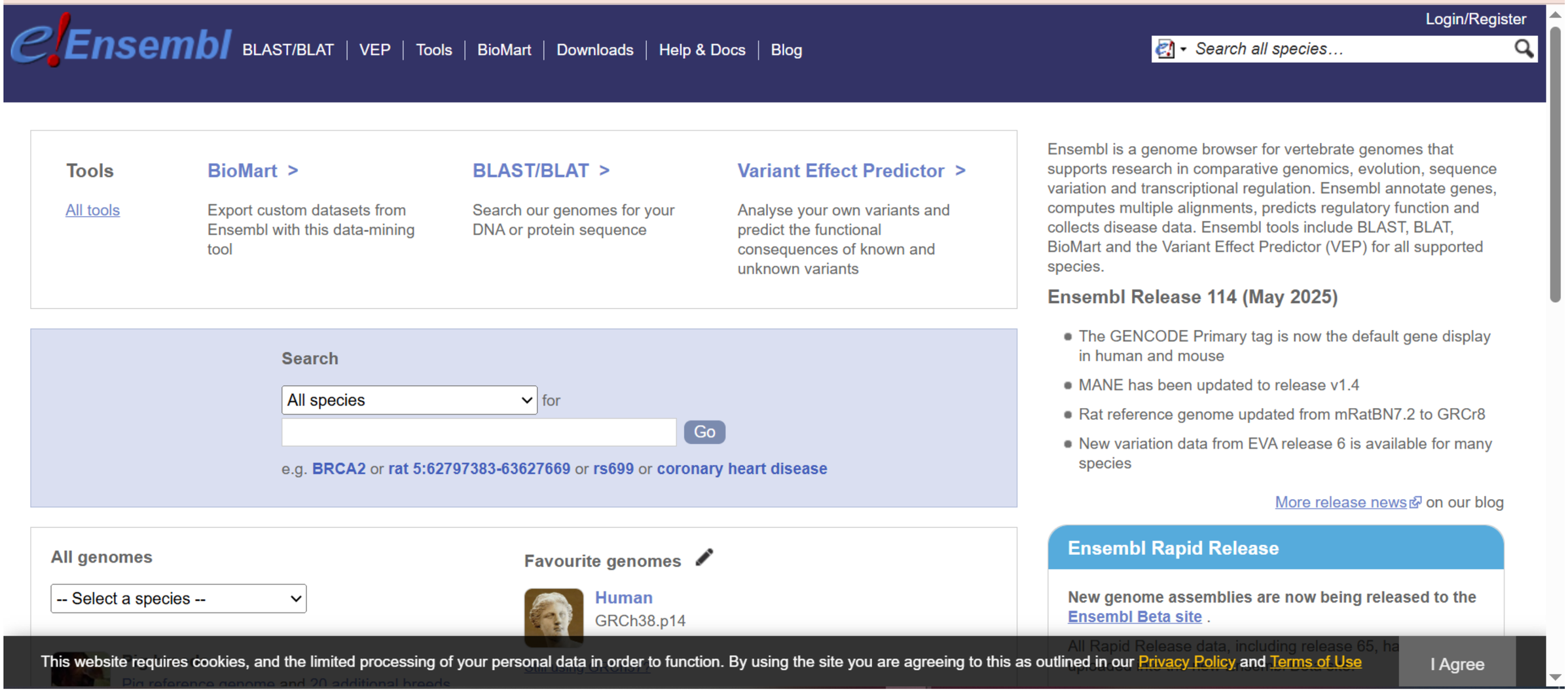
JSON | TSV | Search

	Cases	Project	Data Category	Data Format	File Size	Annotations
	1	HCMI-CMDC	DNA Methylation	IDAT	13.68 MB	0
a_fusion.bedpe	1	HCMI-CMDC	Structural Variation	BEDPE	9.55 kB	0
Tect2.somatic_annotation.vcf.gz	1	HCMI-CMDC	Simple Nucleotide Variation	VCF	2.44 MB	0
Tect2.somatic_annotation.vcf.gz	1	HCMI-CMDC	Simple Nucleotide Variation	VCF	7.1 MB	0
.maf.gz	1	HCMI-CMDC	Simple Nucleotide Variation	MAF	5.52 MB	0
	1	HCMI-CMDC	Sequencing Reads	BAM	64.42 GB	0

Ensembl Genome Browser

- Developed by **EMBL-EBI** and **Sanger Institute.**
- **Annotation** of human, mouse, 100+ vertebrate genomes.
- Provides **reference genomes**, **gene models**, **regulatory features** and **variation data**.
- Link: https://www.ensembl.org/index.html

Ensembl – Homepage



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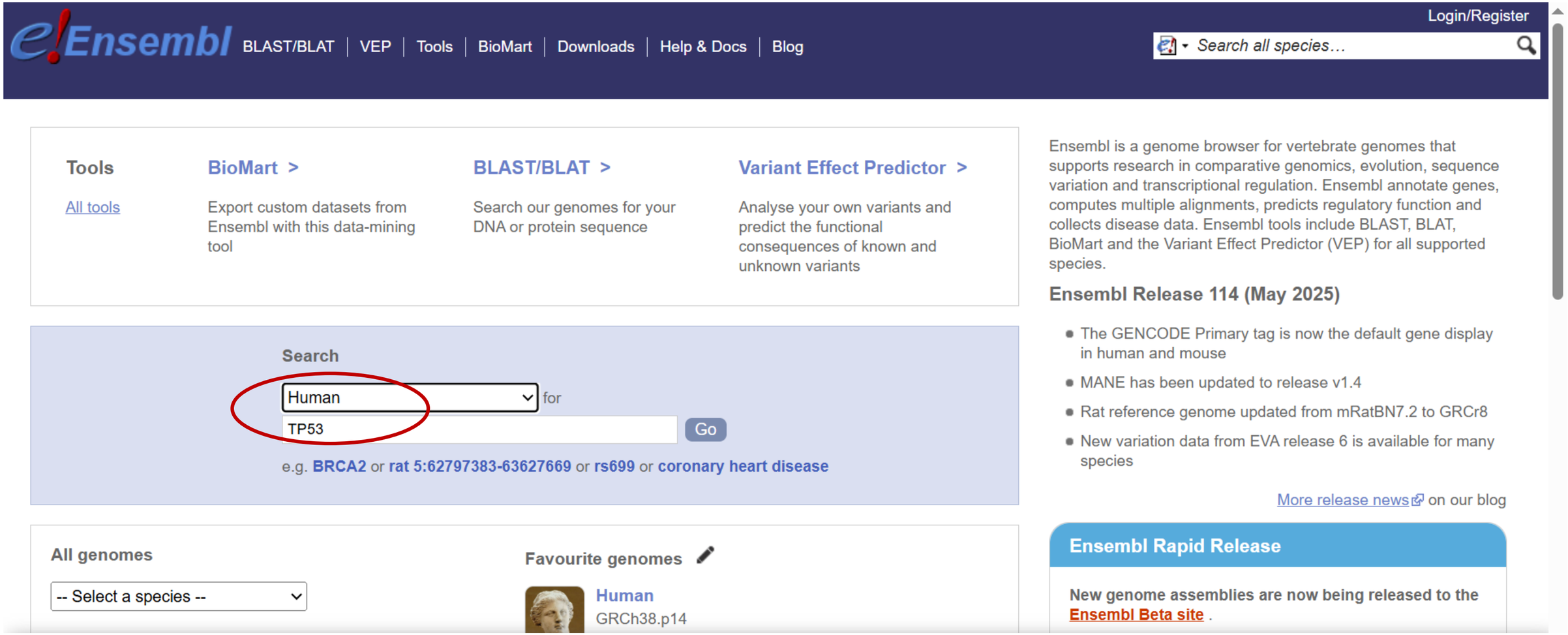

Ensembl – Search for a gene (TP53)

Human (GRCh38.p14) ▾

Location: 17:7,661,779-7,687,546 | Gene: TP53

Gene-based displays
- Summary
  - Splice variants
  - Transcript comparison
  - Gene alleles
- Sequence
  - Secondary Structure
- Comparative Genomics
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- Supporting evidence
- ID History
  - Gene history

Gene: TP53 ENSG00000141510

Description	tumor protein p53 [Source:HGNC Symbol;Acc:HGNC:11998]
Gene Synonyms	LFS1, P53
Location	Chromosome 17: 7,661,779-7,687,546 reverse strand. GRCh38:CM000679.2
About this gene	This gene has 33 transcripts (splice variants), 252 orthologues, 2 paralogues and is associated with 445 phenotypes.
Transcripts	Show transcript table

Summary

Name	TP53 (HGNC Symbol)
MANE	This gene contains MANE Select ENST00000269305, ENSP00000269305
UniProtKB	This gene has proteins that correspond to the following UniProtKB identifiers: P04637
RefSeq	This Ensembl/Gencode gene contains transcript(s) for which we have selected identical RefSeq transcript(s). If there are other RefSeq transcripts available they will be in the External references table
CCDS	This gene is a member of the Human CCDS set: CCDS11118.1, CCDS45605.1, CCDS45606.1, CCDS73963.1, CCDS73964.1, CCDS73965.1, CCDS73966.1, CCDS73967.1, CCDS73968.1, CCDS73969.1, CCDS73970.1, CCDS73971.1
LRG	LRG_321 provides a stable genomic reference framework for describing sequence variants for this gene
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Ensembl – Key Features

- **Gene trees (comparative genomics):** Illustrates evolutionary relationships between orthologs and paralogs.
- **Variant Effect Predictor (VEP):** Predicts functional effects of variants on genes and proteins.
- **REST API** for automated data access: Enables large-scale or customized queries via scripts, useful for reproducible research.

UCSC Genome Browser

- Developed by University of California, Santa Cruz.
- **Visual interface** for **genome navigation**.
- Integrates **multiple data tracks** over reference genomes including genes, methylation, GWAS hits.

UCSC Genome Browser

- Supports human, mouse and others.
- Custom track upload and visualization.
- Real-time queries and **LiftOver** tools.
- Link: https://genome.ucsc.edu/

UCSC Genome Browser – Homepage

UNIVERSITY OF CALIFORNIA SANTA CRUZ Genomics Institute

UCSC Genome Browser

Genomes | Genome Browser | Tools | Mirrors | Downloads | My Data | Projects | Help | About Us

Try our new clinical tutorial!

Search genes, data, help docs and more... Search

Tools

- **Genome Browser** - Interactively visualize genomic data
- **BLAT** - Rapidly align sequences to the genome
- **In-Silico PCR** - Rapidly align PCR primer pairs to the genome
- **Table Browser** - Download and filter data from the Genome Browser
- **LiftOver** - Convert genome coordinates between assemblies
- **REST API** - Returns data requested in JSON format

hg38

hg19

News

May 21, 2025 - **Varaico variants track available for human (hg38 & hg19)**

May 19, 2025 - **New GENCODE "knownGene" V48 for human (hg38/hg19) and VM**

May 14, 2025 - **New Exon Relevance RTS and gnomAD pext track for hg38**

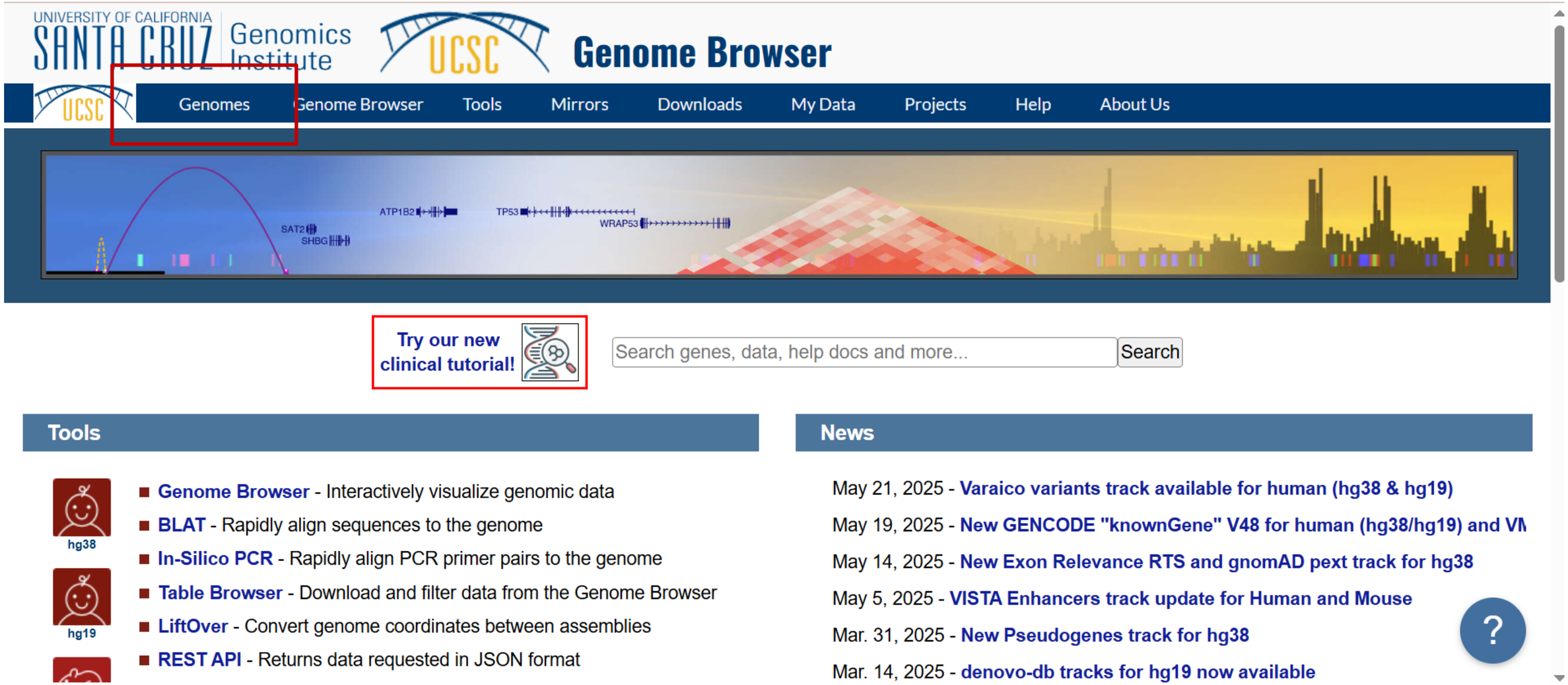
May 5, 2025 - **VISTA Enhancers track update for Human and Mouse**

Mar. 31, 2025 - **New Pseudogenes track for hg38**

Mar. 14, 2025 - **denovo-db tracks for hg19 now available**

?

UCSC Genome Browser – ‘Genomes’



UCSC Genome Browser – ‘Genomes’

Genomes | Genome Browser | Tools | Mirrors | Downloads | My Data | Projects | Help | About Us

Browse/Select Species

POPULAR SPECIES

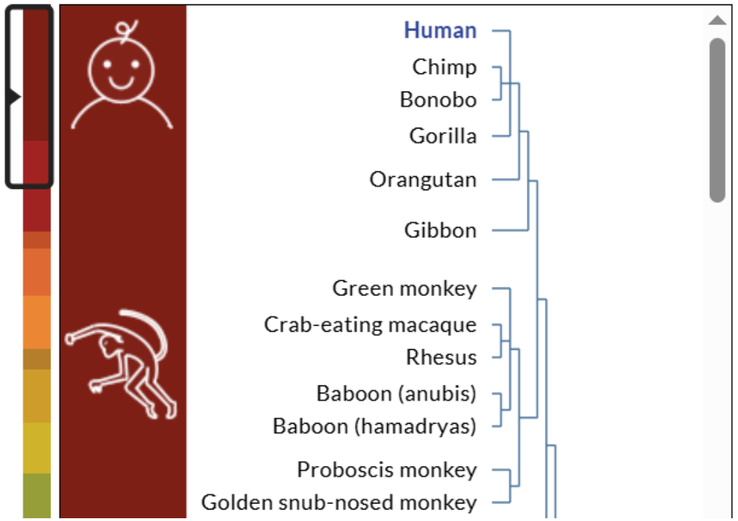
Human | Mouse | Rat | Zebrafish | Fruitfly | Worm | Yeast

Search through thousands of genome browsers

Enter species, common name or assembly ID

Unable to find a genome? Send us a request.

UCSC SPECIES TREE AND CONNECTED ASSEMBLY HUBS



Find Position

Human Assembly

Dec. 2013 (GRCh38/hg38)

GO

Position/Search Term

Enter position, gene symbol or search terms

Current position: chr17:7,673,803-7,673,803

Human Genome Browser - hg38 assembly | View sequences

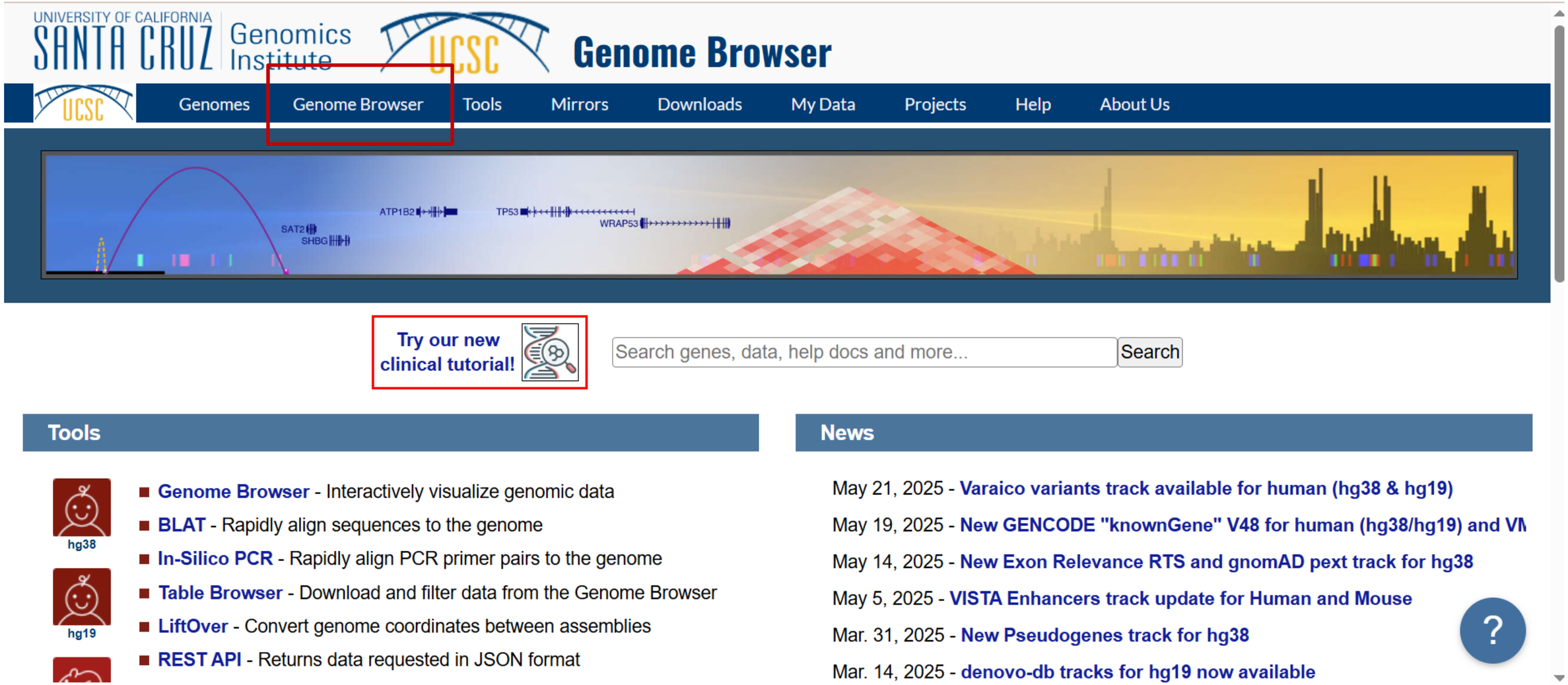
UCSC Genome Browser assembly ID: hg38
Sequencing/Assembly provider ID: Genome Reference Consortium Human GRCh38.p14 (GCA_000001405.29)
Assembly date: Dec. 2013 initial release; June 2022 patch release 14
Assembly accession: GCA_000001405.29
NCBI Genome ID: 51 (Homo sapiens (human))
NCBI Assembly ID: GCF_000001405.40 (GRCh38.p14, GCA_000001405.29)
BioProject ID: PRJNA31257

Homo sapiens
(Graphic courtesy of CBSE)

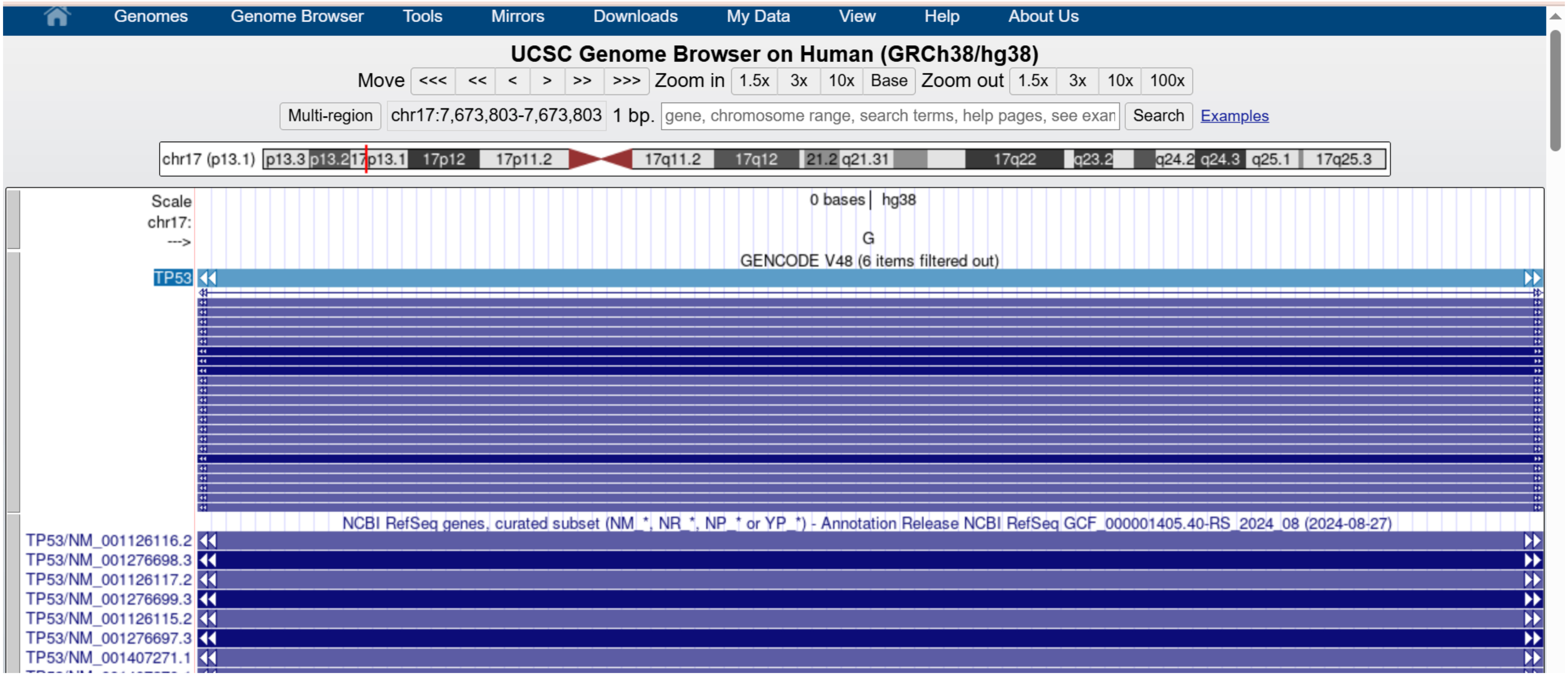
Search the assembly:

- **By position or search term:** Use the "position or search term" box to find areas of the genome associated with many different attributes, such as a specific chromosomal coordinate range; mRNA, EST, or STS marker names; or keywords from the GenBank description of an mRNA. More information, including sample queries.

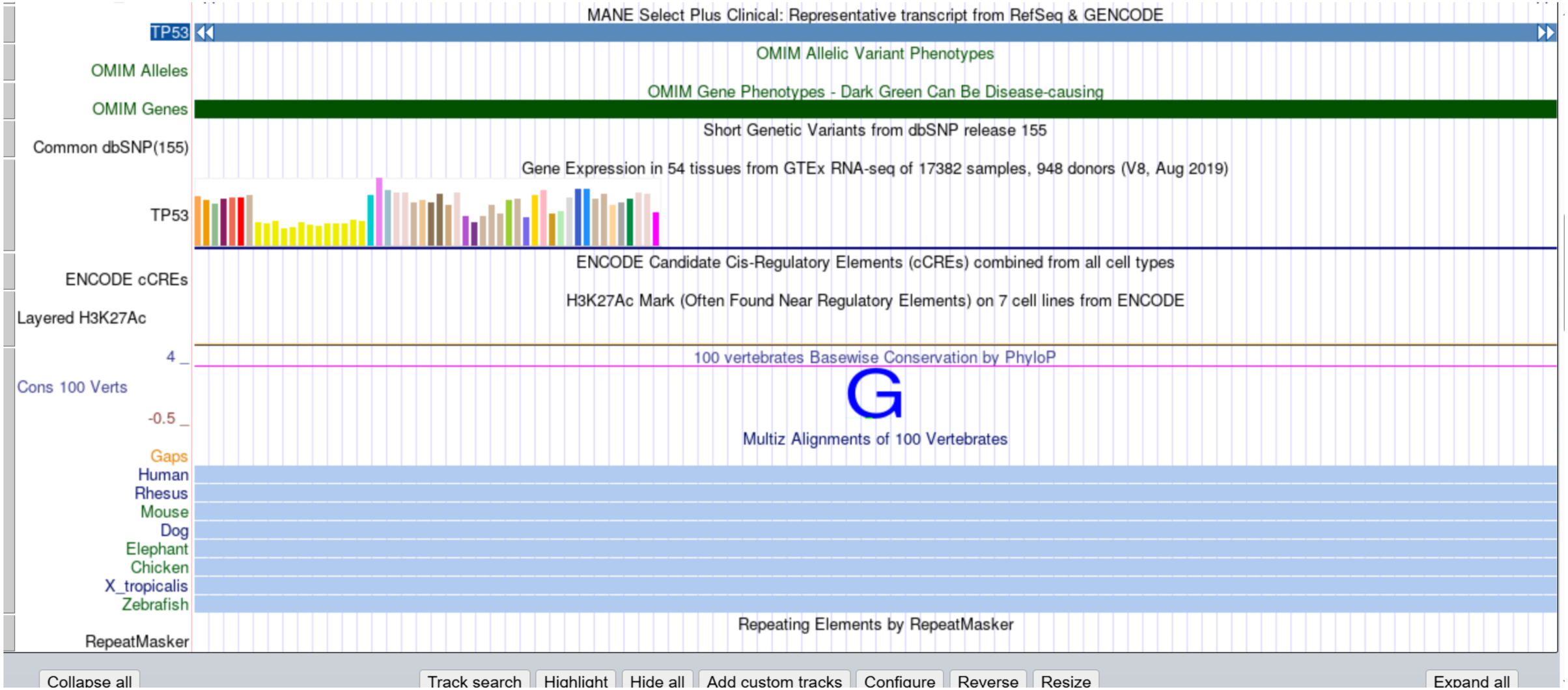
UCSC Genome Browser – ‘Genome Browser’



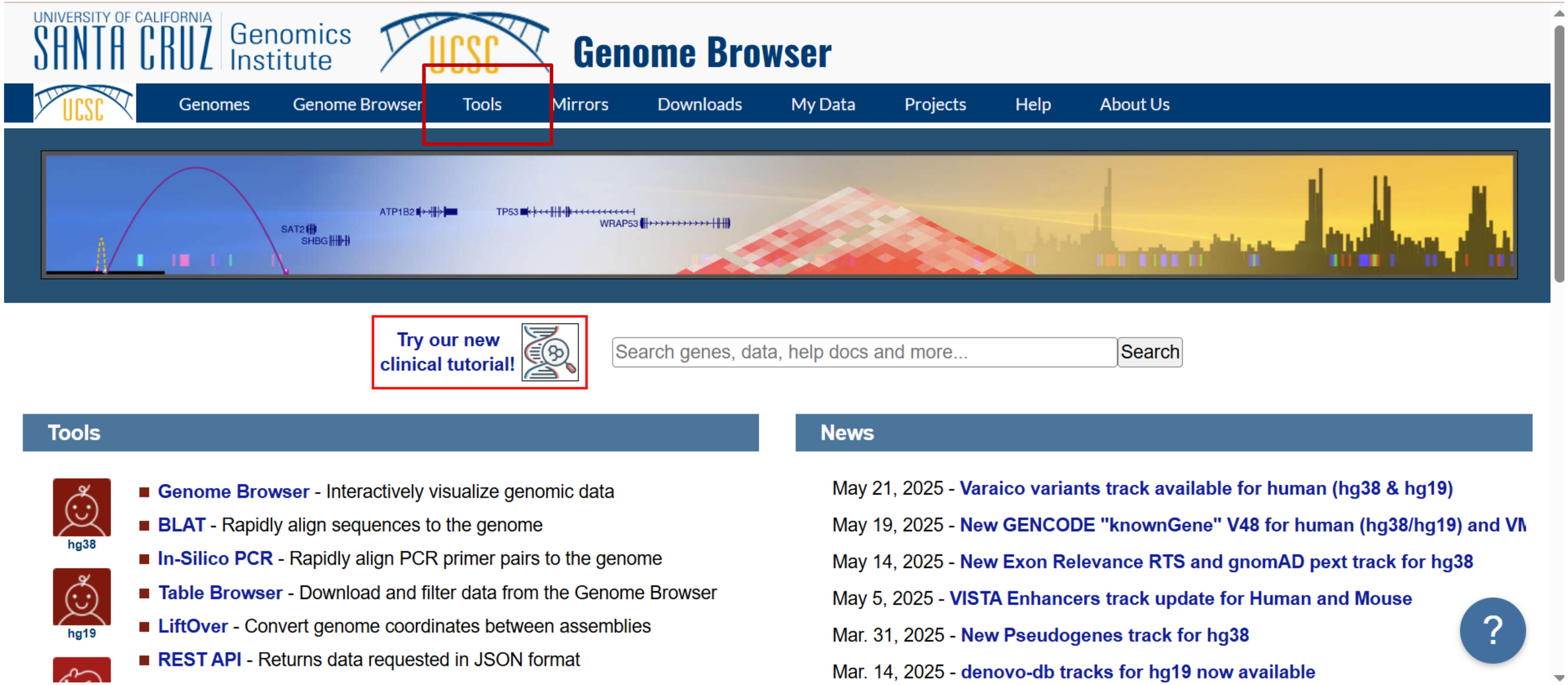
UCSC Genome Browser – ‘Genome Browser’



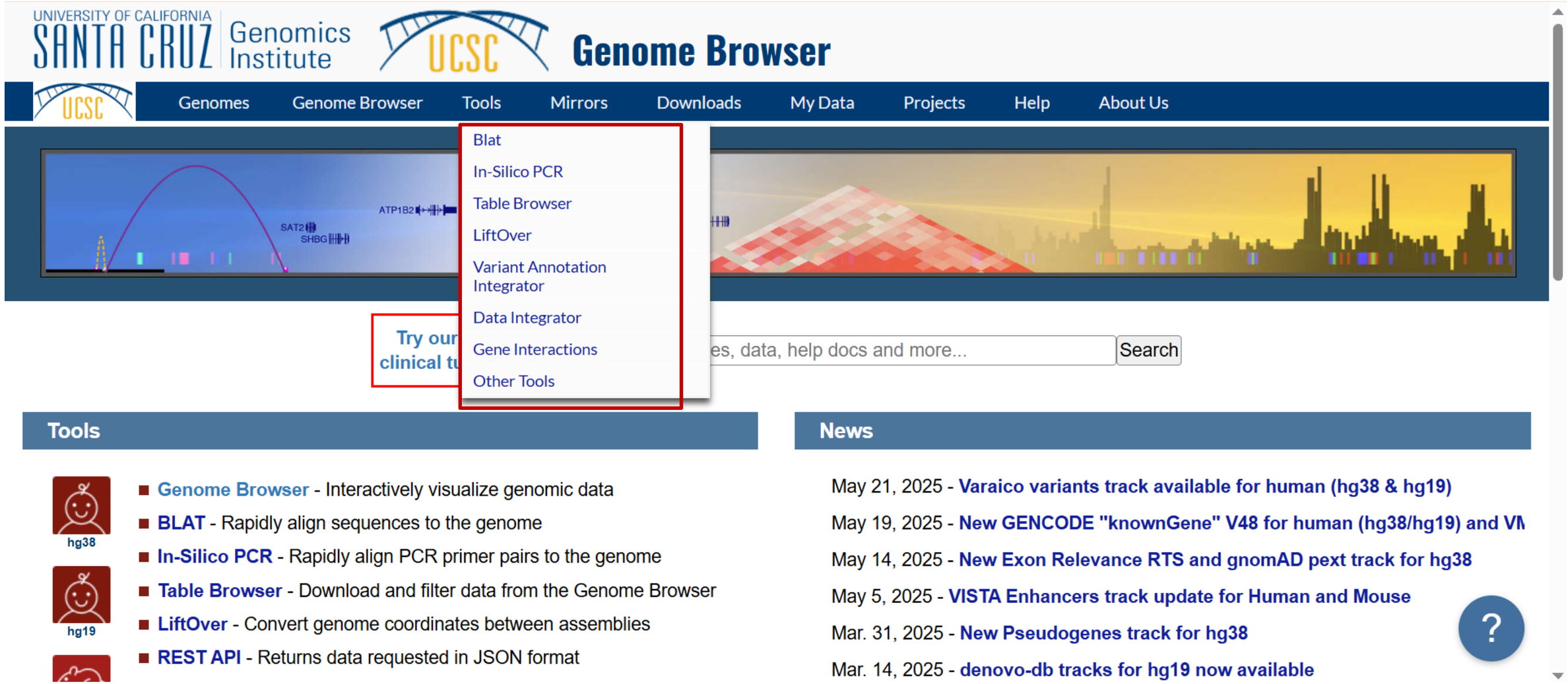
UCSC Genome Browser – 'Genome Browser'



UCSC Genome Browser – 'Tools'

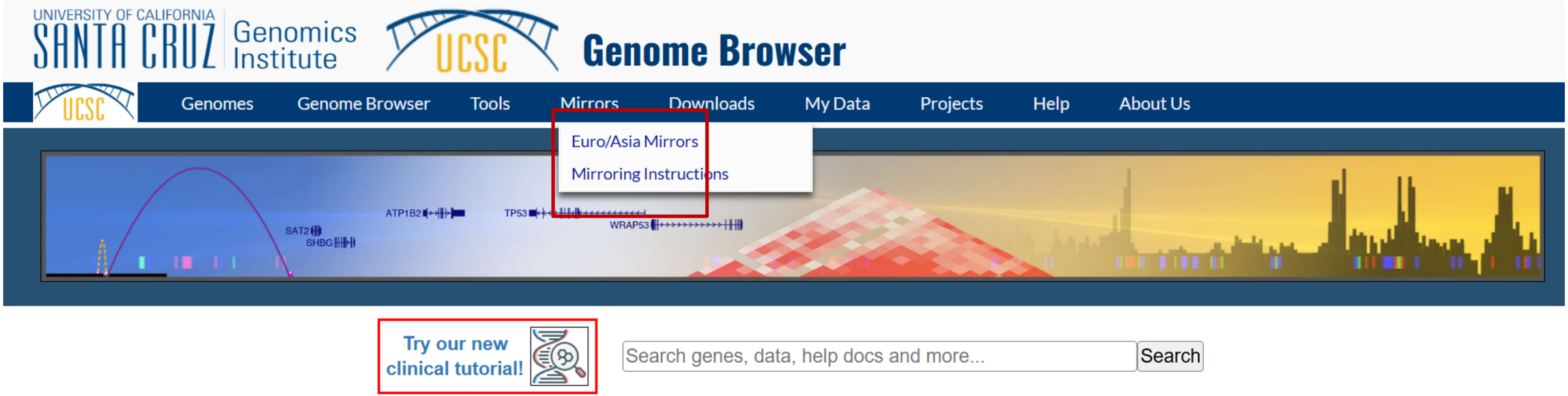


UCSC Genome Browser – 'Tools'



UCSC Genome Browser – 'Mirrors'

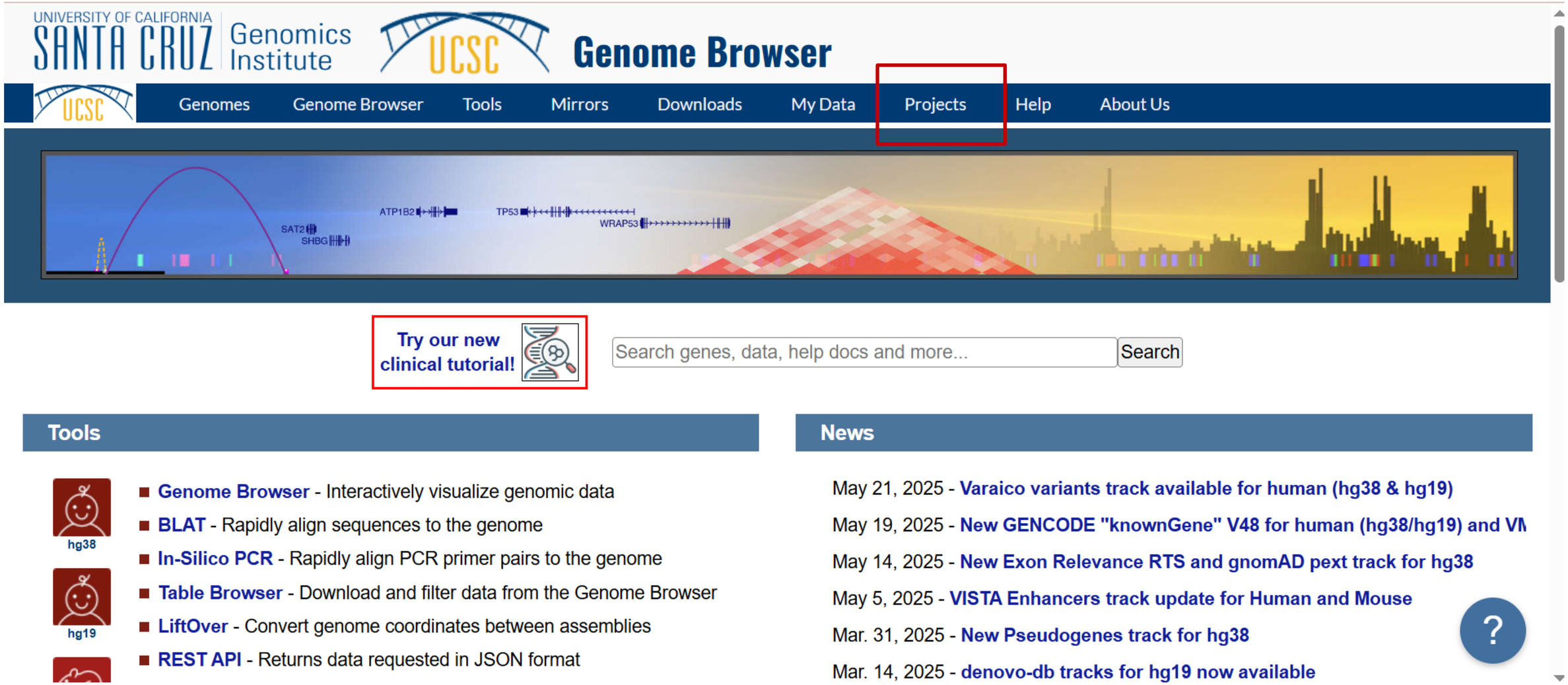
UCSC Genome Browser – 'Mirrors'



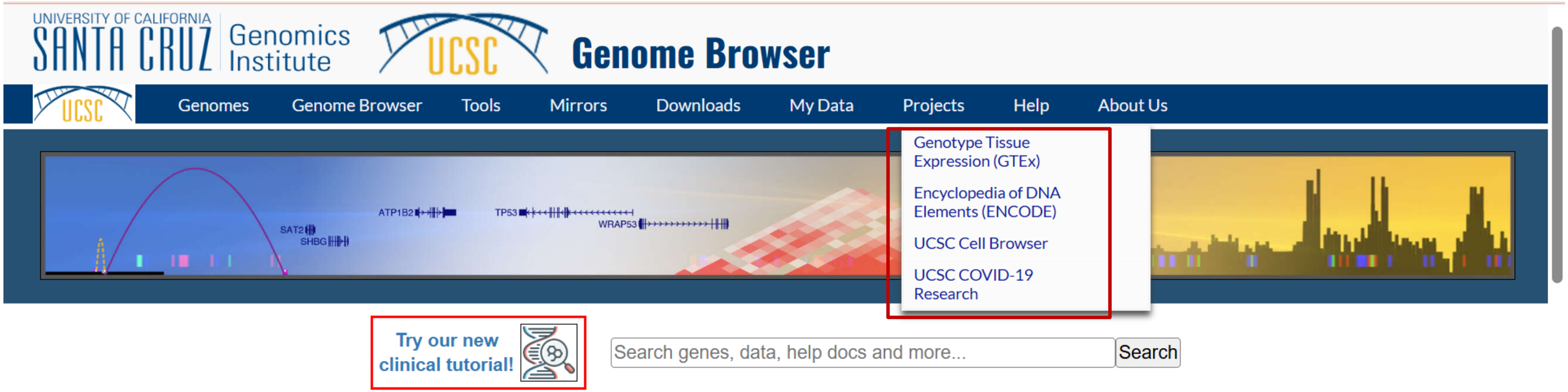
UCSC Genome Browser – 'Downloads'

UCSC Genome Browser – ‘My Data’

UCSC Genome Browser – 'Projects'



UCSC Genome Browser – 'Projects'



Tools

- **Genome Browser** - Interactively visualize genomic data
- **BLAT** - Rapidly align sequences to the genome
- **In-Silico PCR** - Rapidly align PCR primer pairs to the genome
- **Table Browser** - Download and filter data from the Genome Browser
- **LiftOver** - Convert genome coordinates between assemblies
- **REST API** - Returns data requested in JSON format

hg38

hg19

News

May 21, 2025 - **Varaico variants track available for human (hg38 & hg19)**

May 19, 2025 - **New GENCODE "knownGene" V48 for human (hg38/hg19) and VM**

May 14, 2025 - **New Exon Relevance RTS and gnomAD pext track for hg38**

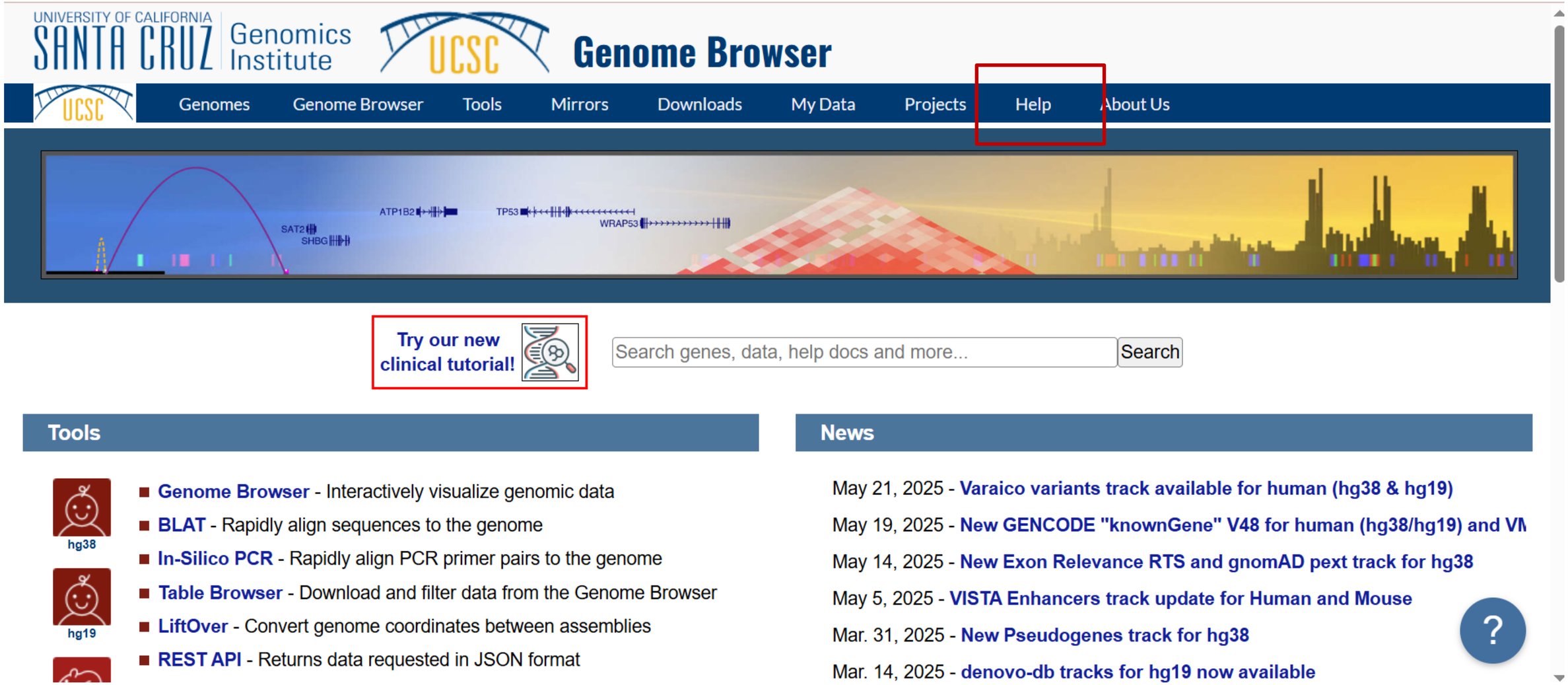
May 5, 2025 - **VISTA Enhancers track update for Human and Mouse**

Mar. 31, 2025 - **New Pseudogenes track for hg38**

Mar. 14, 2025 - **denovo-db tracks for hg19 now available**

?

UCSC Genome Browser – 'Help'



Summary Table

Database	Best For	Tools/Languages
TCGA	Cancer multi-omics + clinical	R, Python, GDC
Ensembl	Genes, orthologs, variants	VEP, REST API
UCSC	Visualizing integrated data	LiftOver, BLAT

Practical Demo…

Using the TCGA data portal:

1. **Search Criteria**
   1. **Disease**: Liver cancer
   2. **Primary site**: *Adenocarcinoma, NOS*
   3. **Experimental strategy**: *RNA-Seq*
2. **Using your search results, answer the following:**
   1. a. How many **mutated genes** are displayed?
   2. b. How many **total mutations** are present?
   3. c. Which gene has the **highest number of mutations**?
   4. d. What is the **mutation frequency** of this gene in liver cancer (in percentage)?
   5. e. What is the **amino acid change** and at what **position** did it occur? *(e.g., Alanine (A) → Arginine (R) at position 175)*

Practical Demo...

3. Biological Impact (Advanced Task)

Using **TCGA's 'Impact' column** or by cross-checking the mutation in the **Ensembl Variant Effect Predictor (VEP)** or **PolyPhen**:

•**f.** What is the predicted effect of this mutation? *(*e.g., "probably damaging", "deleterious", "moderate impact"*)*

•**g.** How might this mutation affect the protein's structure or function? *(*Briefly explain in 2–3 sentences.*)*

Practical Tips for Using Databases

- Always note genome build (e.g., hg19 vs hg38).
- Use filters to avoid downloading massive datasets.
- Read documentation, each tool has quirks

Further Reading/ References

- **UCSC** Video Tutorial: https://genome.ucsc.edu/cgi-bin/hgTracks?db=hg38&startClinical=true
- **UCSC** Video Tutorial: https://genome.ucsc.edu/training/
- **TCGA** Video Tutorial: https://docs.gdc.cancer.gov/Data_Portal/Users_Guide/Video_Tutorials/
- **Ensembl** Video Tutorial: https://www.ensembl.org/info/website/tutorials/index.html

Conclusion

- Biological databases are essential for computational biology analysis. Learn to navigate and cross-reference them because **data literacy is power** in computational biology.

Coming up next…

Obtaining data from databases:

- ✓ CNVs
- ✓ RNA-Seq
- ✓ Clinical data